# Library

Stronger buffers cost input cap but drive with lower resistance

# The idea

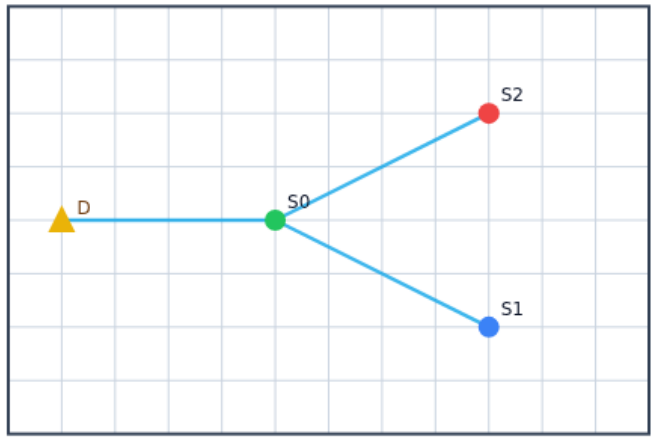
- BUF_X1, BUF_X2, INV_X1 each publish cin, rout, delay, invert
- Van Ginneken tries each non-inverting cell unless polarity mode allows INV

# Three cells

BUFFER LIBRARY

## Three cells

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**Explanation**

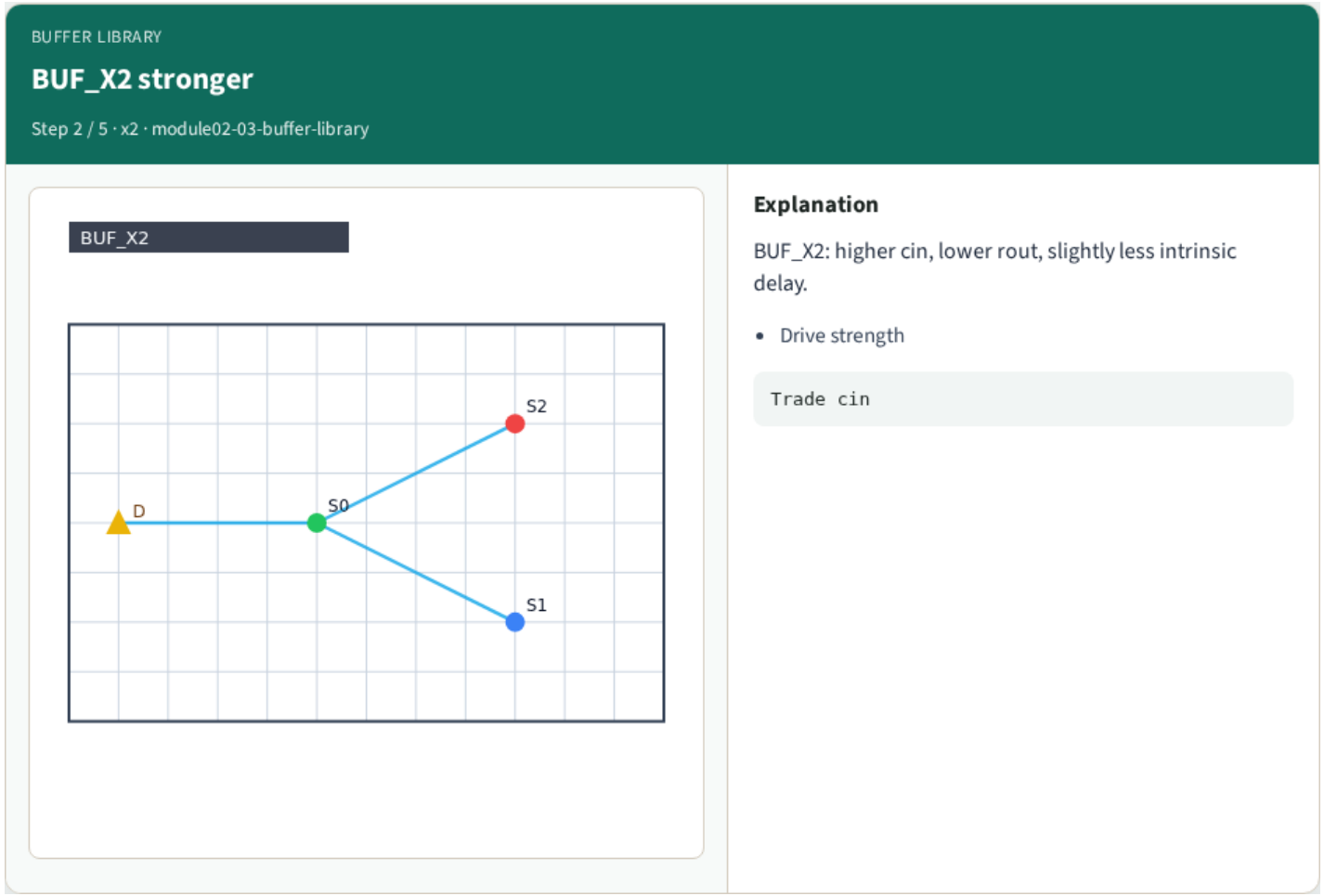
Library has BUF_X1, BUF_X2, INV_X1.

- cin/rout/delay

```
lib=3
```

Three cells

# BUF_X2 stronger



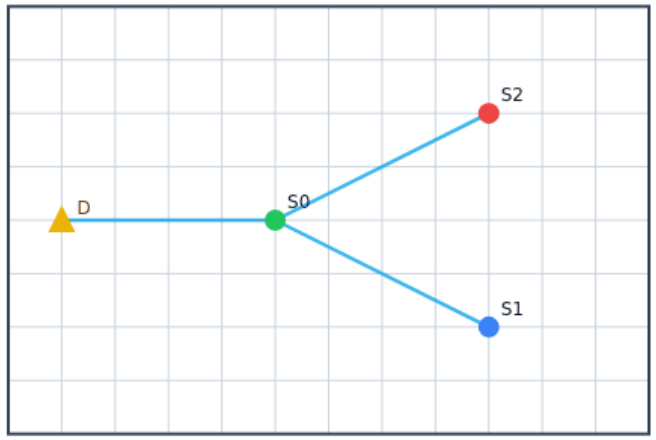

BUF_X2 stronger

# Inverter

BUFFER LIBRARY

## Inverter

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**Explanation**

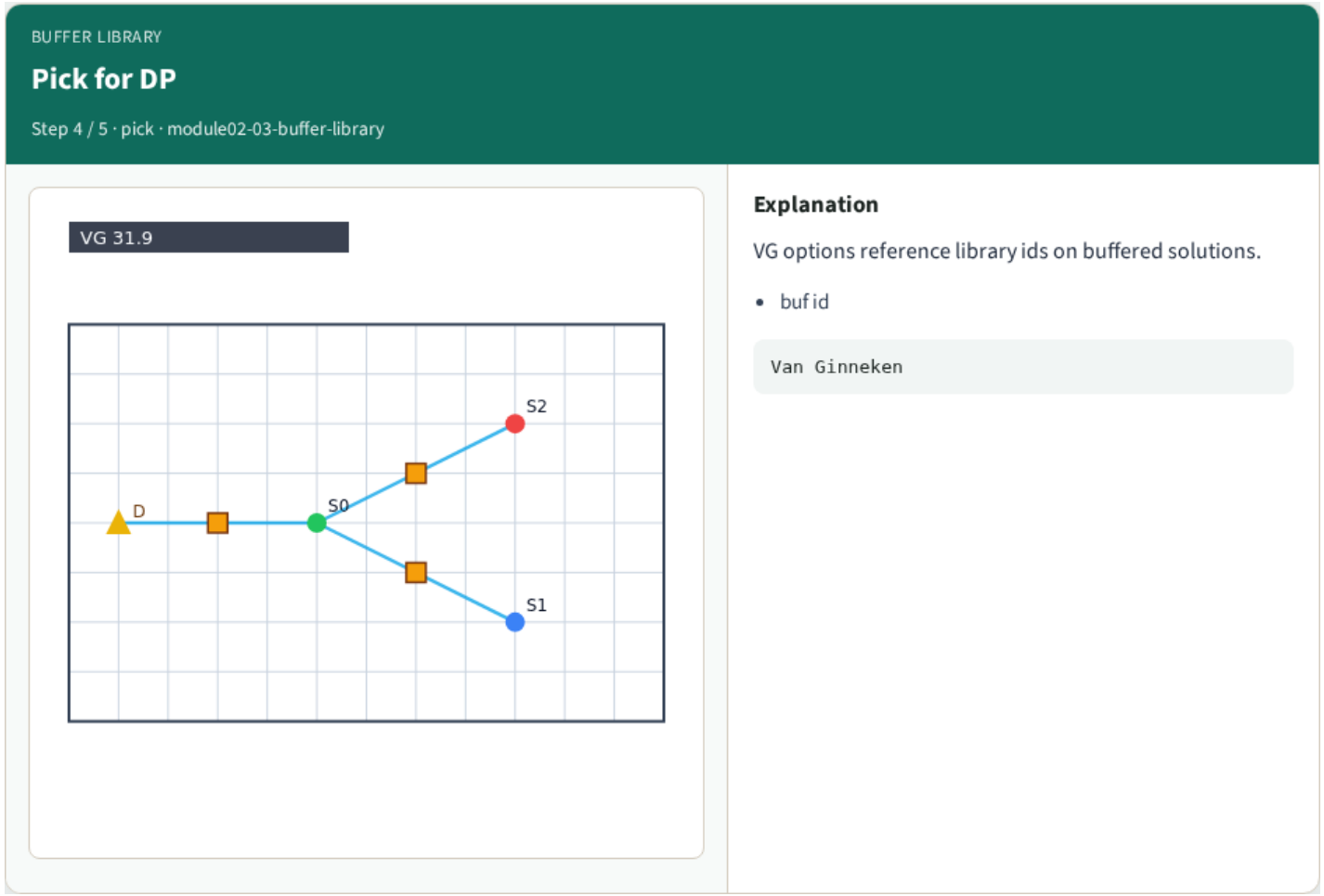
INV_X1 flips polarity — used in polarity labs.

- invert=true

```
Purple later
```

Inverter

# Pick for DP



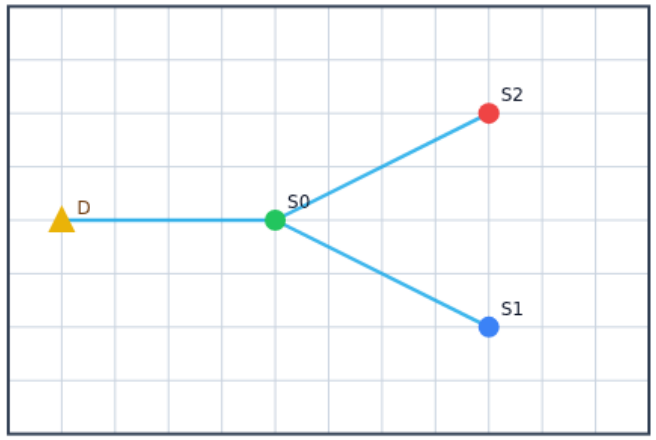

Pick for DP

# Area trade

BUFFER LIBRARY

## Area trade

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**Explanation**

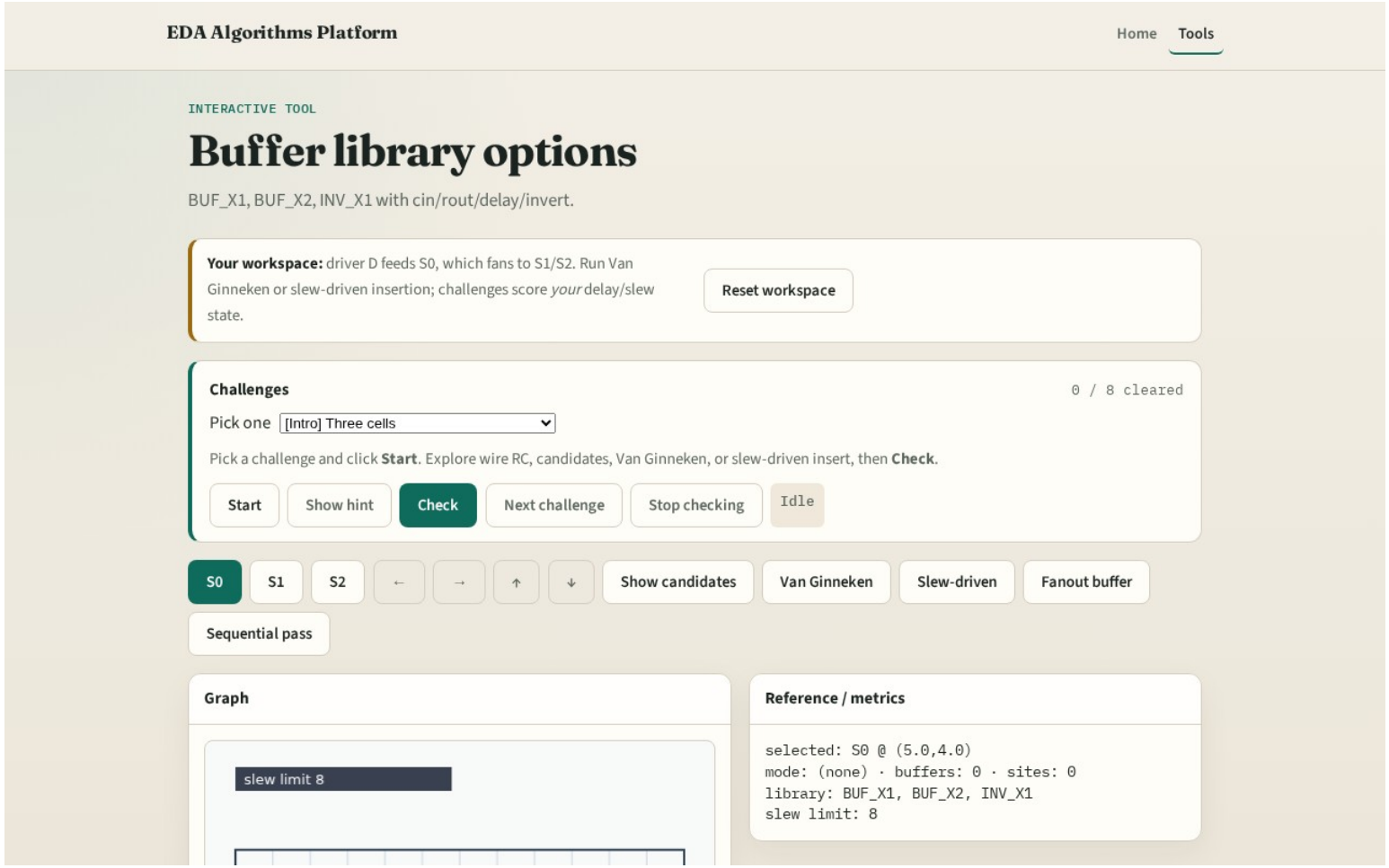
Real flows also score area; this toy scores delay/slew only.

- Toy scope

```
Next: VG
```

Area trade

# Browser lab track




Browser lab starter

# Implement track

- Print `data['buffer_library']`
- Assert GOLDENS.libCount == 3

# Pitfalls

- Ignoring invert flag
- Treating rout as delay
- Using cells not in the library

# Your turn

- Next: Van Ginneken dynamic programming lite